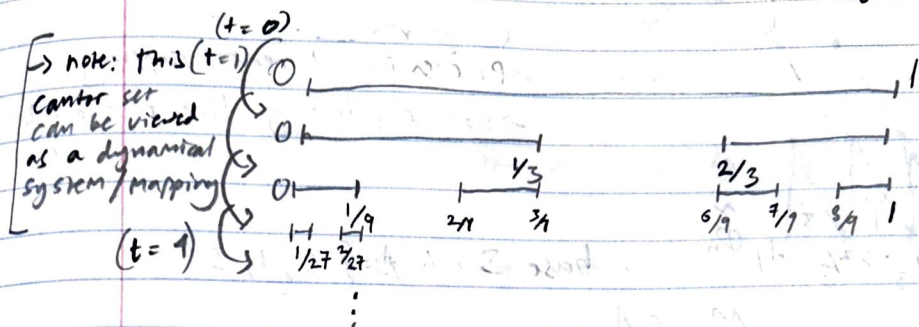


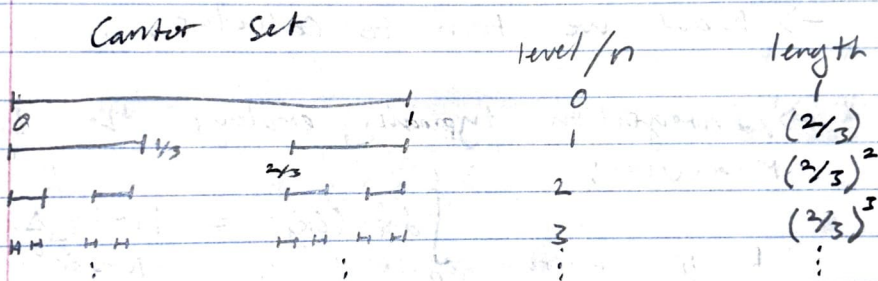
Lecture 11-06-2025

"Fractals & Ischaemicals"

11.2 Cantor Set (Cardinality of infinities)



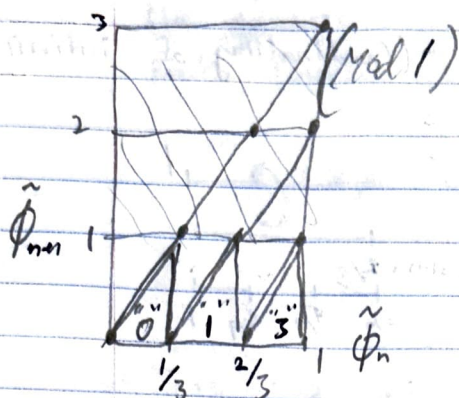
- eventually, the points become infinitesimal: "Cantor dust"
- but how do you measure things that have no size, i.e., this set in the infinite limit



- from the perspective of each $n+1$ th layer, the n th layer is size 1 compared to $1/3$: the layers are SELF-SIMILAR

- we begin to use the concept that if you resolve the system in chaotic dynamics you lose self-similarity
- this is a non-integer dimensional dynamical system
- We make this into an explicit dynamical system

→ Bernoulli Map:



$$x_{n+1} = 3x_n \text{ Mod } 1$$

→ piecewise linear mapping
→ not explicitly linear though

$$\text{base } 3 : A = \{0, 1, 2\}$$

$$M_n \in A$$

$$\tilde{\phi} = \frac{M_1}{3} + \frac{M_2}{3^2} + \frac{M_3}{3^3} + \frac{M_4}{3^4} \dots$$

$$= 0.M_1M_2M_3M_4\dots$$

→ Now we turn to calculus...

→ Integration typically employs the Lebesgue measure:

$$\int_0^1 dx f(x) = \lim_{N \rightarrow \infty} \sum_{i=1}^N \Delta x_i f_i$$

→ But this technique breaks down when trying to integrate something like the Cantor Map, since there are many "holes"

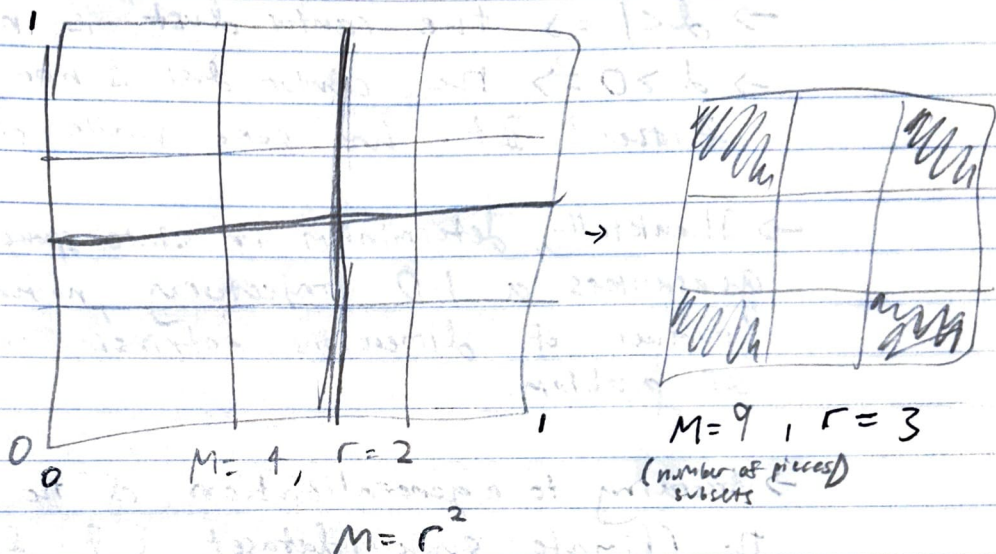
⇒ We turn back to renormalization: scale the 'size' of each (successively smaller) interval

→ Hausdorff Dimension / Measures

$$S_n(\tau) = \sum_i^n \Delta_i^\tau$$

→ A critical dimension τ_H occurs when $S_n = 1$
 → optimal cover

→ Now, consider the Cantor set in 2D:



→ For an n -dimensional space, $M=r^n$ for the Cantor set

→ Consider

$$\frac{\log M}{\log r} = \frac{\log(\# \text{ pieces})}{\log(\text{size of piece})}$$

$$\Rightarrow \log M = d \log r$$

$$\Rightarrow d = \frac{\log M}{\log r} = \text{constant}$$

→ based on the notion of similarity dimension

→ For the Cantor set, this yields:

Similarity Dimension $d = \frac{\log 2^n}{\log 3^n} = \frac{\log 2}{\log 3} \approx 0.63$

→ This dimension is not an integer!

→ $d < 1 \Rightarrow$ the 'Cantor dust' is not a line!

→ $d > 0 \Rightarrow$ the Cantor dust is not a point, ^{interval} either! It has some finite size

→ Thankfully, determinism in state space produces/guarantees a 1-D trajectory, no matter the number of dimensions intrinsic to the d or problem

→ According to a generalization of the above, the Climate system/dataset is 7-dimensional

→ Ruelle (another friend of Predrag) ↗

→ Grassberger & Procaccia (more friends) define the 'correlation dimension':

Phase space

